PHYSICAL CHEMISTRY CUMULATIVE EXAM

January 26, 2001

The wavefunction of a 1s-electron in a hydrogen atom is $\psi = N\exp(-r/a_o)$, where N is a constant, $a_o = 52.9$ pm, and r is the distance from the nucleus.

a)Normalize ψ.

b)At which point in space is it most likely to find the electron? Present a mathematical justification for your answer.

c)What is the probability to find the electron inside a small volume of magnitude 1.0 pm^3 located at the Bohr radius, $r=a_o$?

d)Calculate the average value of distance of an electron from the nucleus in the hydrogen atom 1s state.

Each question 25%.

Even powers of x	Odd powers of x
1. $\int_{-\infty}^{\infty} x^{2n}e^{-ax^2}\,dx = 2\int_0^{\infty} x^{2n}e^{-ax^2}\,dx$	4. $\int_{-\infty}^{\infty} x^{2n+1}e^{-ax^2}\,dx = 0$
2. $\int_0^{\infty} e^{-ax^2}\,dx = \frac{\pi^{1/2}}{2a^{1/2}}$	5. $\int_0^{\infty} xe^{-ax^2}\,dx = \frac{1}{2a}$
3. $\int_0^{\infty} x^{2n}e^{-ax^2}\,dx = \frac{(2n)!\,\pi^{1/2}}{2^{2n+1}n!\,a^{n+1/2}}$	6. $\int_0^{\infty} x^{2n+1}e^{-ax^2}\,dx = \frac{n!}{2a^{n+1}}$

where $a > 0$ and $n = 0, 1, 2, \ldots$

$$\int_0^{\infty} x^n e^{-ax}\,dx = n!/a^{n+1}$$